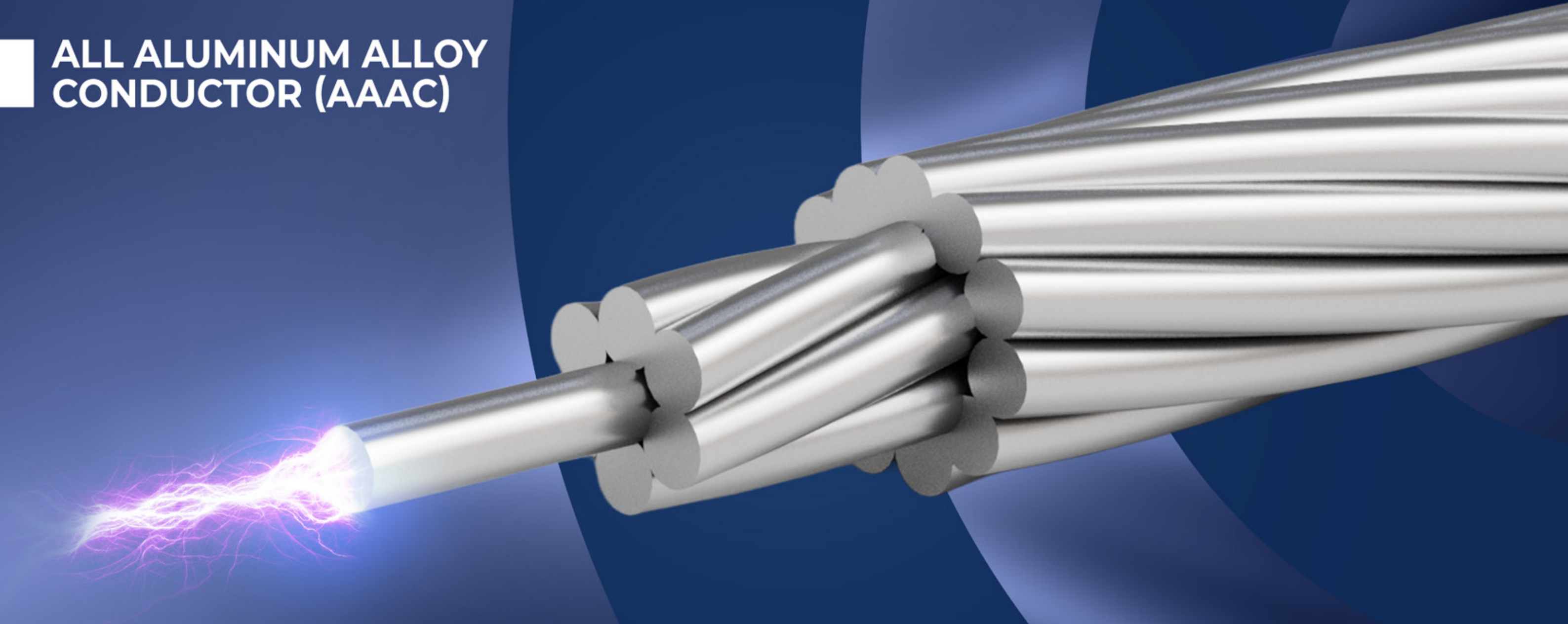


**ALL ALUMINUM ALLOY
CONDUCTOR (AAAC)**



All Aluminum Conductors
(AAAC)

Description

All Aluminum Alloy wires , stranded in successive layers, in opposite direction to form the Aluminum stranded AAAC conductor.

Application

All Aluminum-Alloy Conductors (AAAC) can be used in Medium, High and Extra-High voltage transmission lines. AAAC offers better sag performance due to the high strength-to-weight ratio provided by the aluminum-alloy. In addition, AAAC provides a higher corrosion resistance than ACSR conductors. They are also used a messenger to support overhead electrical cables.

Technical Data

Relevant Standard	BS EN 50182, DIN 48201-6, IEC 61089, ASTM B 399
Conductor	Plain bare All Aluminum Alloy
Minimum bending Radius	25 x outer diameter
Packing Condition	Our standard length on Wooden Drum / or according to customer request

Product Code	Cross Sectional Area mm²	Minimum Number of Wires No. x mm	Max. D.C. Conductor Resistance at 20 °C Ω/km	Approx. Overall Diameter mm²	Minimum Breaking Load KN	Approx. Weight Kg/km	Rated Current Amps
A- BS EN 50182 (GERMANY) & DIN 48201-6							
PO-LT11-00-01-U0	16	7 x 1.70	2.0710	5.10	4.69	43.4	105
PO-LT12-00-01-U0	25	7 x 2.10	1.3566	6.30	7.15	66.2	135
PO-LT13-00-01-U0	35	7 x 2.50	0.9572	7.50	10.14	93.8	170
PO-LT14-00-01-U0	50	7 x 3.00	0.6647	9.00	14.60	135.1	210
PO-LT14-00-01-U0	50	19 x 1.80	0.6841	9.00	14.26	132.7	210
PO-LT15-00-01-U0	70	19 x 2.10	0.5026	10.50	19.41	180.7	255
PO-LT16-00-01-U0	95	19 x 2.50	0.3546	12.50	27.51	256.0	320
PO-LT17-00-01-U0	120	19 x 2.80	0.2827	14.00	34.51	321.2	365
PO-LT18-00-01-U0	150	37 x 2.25	0.2256	15.80	43.40	405.3	425
PO-LT19-00-01-U0	185	37 x 2.50	0.1827	17.50	53.58	500.3	490
PO-LT20-00-01-U0	240	61 x 2.25	0.1373	20.30	71.55	670.3	585
PO-LT21-00-01-U0	300	61 x 2.50	0.1112	22.50	88.33	827.5	670
PO-LT22-00-01-U0	400	61 x 2.89	0.0832	26.00	118.04	1105.9	810
PO-LT23-00-01-U0	500	61 x 3.23	0.0666	29.10	147.45	1381.4	930



Product Code	Cross Sectional Area mm²	Minimum Number of Wires No. x mm	Max. D.C. Conductor Resistance at 20 °C Ω/km	Approx. Overall Diameter mm²	Minimum Breaking Load KN	Approx. Weight Kg/km	Rated Current Amps
B - BS EN 50182 (United Kingdom)							
BOX	18.8	7 x 1.85	1.7480	5.55	5.55	51.4	112
ACACIA	23.8	7 x 2.08	1.3828	6.24	7.02	64.9	130
ALMOND	30.1	7 x 2.34	1.0926	7.02	8.88	82.2	151
CEDAR	35.5	7 x 2.54	0.9273	7.62	10.46	96.8	168
DEODAR	42.2	7 x 2.77	0.7797	8.31	12.44	115.2	187
FIR	47.8	7 x 2.95	0.6875	8.85	14.11	130.6	202
HAZEL	59.9	7 x 3.30	0.5494	9.90	17.66	163.4	233
PINE	71.6	7 x 3.61	0.4591	10.80	21.14	195.6	261
HOLLY	84.1	7 x 3.91	0.3913	11.70	24.79	229.5	289
WILLOW	89.7	7 x 4.04	0.3665	12.10	26.47	245.0	301
OAK	118.9	7 x 4.65	0.2767	14.00	35.07	324.5	360
MULBERRY	150.9	19 x 3.18	0.2192	15.90	44.52	414.3	420
ASH	180.7	19 x 3.48	0.183	17.40	53.31	496.1	471
ELM	221	19 x 3.76	0.1568	18.80	62.24	579.2	519
POPLAR	239.4	37 x 2.87	0.1387	20.10	70.61	659.4	562
SYCAMORE	303.2	37 x 3.23	0.1095	22.60	89.40	835.2	654
UPAS	362.1	37 x 3.53	0.0917	24.70	106.82	997.5	732
YEW	479	37 x 4.06	0.0693	28.40	141.31	1319.6	874
TOTARA	498.1	37 x 4.14	0.0666	29.00	146.93	1372.1	895
RUBUS	586.9	61 x 3.50	0.0567	31.50	173.13	1622.0	991
SORBUS	659.4	61 x 3.71	0.0505	33.40	194.53	1822.0	1066
ARAUCARIA	821.1	61 x 4.14	0.0406	37.30	242.24	2269.4	1221
REDWOOD	996.2	61 x 4.56	0.0334	41.00	293.88	2753.2	1373

The above data is approximate and subject to manufacturing tolerances

Note:-

The above values of current carrying capacity are calculated due to frequency up to 60 Hz, assuming an ambient temperature of 35 °C , final temperature of 80 °C, wind velocity of 0.6 m/sec, solar radiation of 900 W/m2, emissivity coefficient of 0.6, solar radiation absorption coefficient of 0.5

All Aluminum Alloy Conductors
(AAAC)

Product Code	Cross Sectional Area mm²	Minimum Number of Wires No. x mm	Max. D.C. Conductor Resistance at 20 °C Ω/km	Approx. Overall Diameter mm²	Minimum Breaking Load KN	Approx. Weight Kg/Km	Rated Current Amps
C - BS EN 50182 (France)							
ASTER - 22	22	7 x 2.00	1.4989	6.00	7.15	60.0	126
ASTER - 34.4	34.4	7 x 2.50	0.9593	7.50	11.17	93.8	168
ASTER - 54.6	54.6	7 x 3.15	0.6042	9.45	17.73	148.9	225
ASTER - 75.5	75.5	19 x 2.25	0.4388	11.30	24.55	207.4	278
ASTER - 117	117	19 x 2.80	0.2833	14.00	38.02	321.2	367
ASTER - 148.1	148.1	19 x 3.15	0.2239	15.80	48.12	406.5	427
ASTER - 181.6	181.6	37 x 2.50	0.1831	17.50	59.03	500.3	487
ASTER - 227.8	227.8	37 x 2.80	0.1460	19.60	74.04	627.6	563
ASTER - 288.3	288.3	37 x 3.15	0.1154	22.10	93.71	794.3	655
ASTER - 266.2	266.2	37 x 3.55	0.0908	24.90	115.36	1008.9	764
ASTER - 570.2	570.2	61 x 3.45	0.0585	31.10	185.33	1576.0	1013

Product Code	Cross Sectional Area mm²	Minimum Number of Wires No. x mm	Max. D.C. Conductor Resistance at 20 °C Ω/km	Approx. Overall Diameter mm²	Minimum Breaking Load KN	Approx. Weight Kg/Km	Rated Current Amps
D - IEC 61089 (Aluminum Alloy A2 - Type B)							
A2 - 16	18.4	7 x 1.83	1.7896	5.49	5.43	50.4	150
A2 - 25	28.8	7 x 2.29	1.1453	6.86	8.49	78.7	200
A2 - 40	46	7 x 2.89	0.7158	8.68	13.58	125.9	270
A2 - 63	72.5	7 x 3.63	0.4545	10.90	21.39	198.3	361
A2 - 100	115	19 x 2.78	0.2877	13.90	33.95	316.3	487
A2 - 125	144	19 x 3.10	0.2302	15.50	42.44	395.4	562
A2 - 160	184	19 x 3.51	0.1798	17.60	54.32	506.1	659
A2 - 200	230	19 x 3.93	0.1439	19.60	67.91	632.7	760
A2 - 250	288	19 x 4.39	0.1151	22.00	84.88	790.8	879
A2 - 315	363	37 x 3.53	0.0916	24.70	106.95	998.9	1019
A2 - 400	460	37 x 3.98	0.0721	27.90	135.81	1268.4	1188
A2 - 450	518	37 x 4.22	0.0641	29.60	152.79	1426.9	1282
A2 - 500	575	37 x 4.45	0.0577	31.20	169.76	1585.5	1370
A2 - 560	645	61 x 3.67	0.0516	33.00	190.14	1778.4	1472
A2 - 630	725	61 x 3.89	0.0458	35.00	213.90	2000.7	1585
A2 - 710	817	61 x 4.13	0.0407	37.20	241.07	2254.8	1706
A2 - 800	921	61 x 4.38	0.0361	39.50	271.62	2540.6	1840

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E - IEC 61089 (Aluminum Alloy A3 - Type A)							
A3 - 16	18.6	7 x 1.84	1.7896	5.52	6.04	50.8	151
A3 - 25	29	7 x 2.30	1.1453	6.90	9.44	79.5	200
A3 - 40	46.5	7 x 2.91	0.7158	8.72	15.10	127.1	270
A3 - 63	73.2	7 x 3.65	0.4545	10.90	23.06	200.2	361
A3 - 100	116	19 x 2.79	0.2877	14.00	37.76	319.3	488
A3 - 125	145	19 x 3.12	0.2302	15.60	42.20	399.2	563
A3 - 160	186	19 x 3.53	0.1798	17.60	58.56	511.0	659
A3 - 200	232	19 x 3.95	0.1439	19.70	73.20	638.7	761
A3 - 250	290	19 x 4.41	0.1151	22.10	91.50	798.4	880
A3 - 315	366	37 x 3.55	0.0916	24.80	115.29	1008.4	1020
A3 - 400	465	37 x 4.00	0.0721	28.00	146.40	1280.5	1190
A3 - 450	523	37 x 4.24	0.0641	29.70	164.70	1440.5	1283
A3 - 500	581	37 x 4.47	0.0577	31.30	183.00	1600.6	1371
A3 - 560	651	61 x 3.69	0.0516	33.20	204.96	1795.3	1474
A3 - 630	732	61 x 3.91	0.0458	35.20	230.58	2019.8	1587
A3 - 710	825	61 x 4.15	0.0407	37.30	259.86	2276.2	1708
A3 - 800	930	61 x 4.40	0.0361	39.60	292.80	2564.8	1841

The above data is approximate and subject to manufacturing tolerances

Note:-

The above values of current carrying capacity are calculated due to frequency up to 60 Hz, assuming an ambient temperature of 35 °C , final temperature of 80 °C, wind velocity of 0.6 m/sec, solar radiation of 900 W/m2, emissivity coefficient of 0.6, solar radiation absorption coefficient of 0.5

All Aluminum Alloy Conductors
(AAAC)

Product Code	Cross Sectional Area mm ²	Minimum Number of Wires No. x mm	Max. D.C. Conductor Resistance at 20 °C Ω/km	Approx. Overall Diameter mm ²	Minimum Breaking Load KN	Approx. Weight Kg/Km	Rated Current Amps
F - ASTM B 399							
6	13.2	7 x 1.55	2.53610	4.65	4.18	36.2	91
4	21.1	7 x 1.96	1.58600	5.88	6.69	57.9	122
2	33.5	7 x 2.47	0.99870	7.41	10.65	92.0	164
0	53.5	7 x 3.12	0.62592	9.36	17.00	146.8	221
2 / 0	67.3	7 x 3.50	0.49738	10.50	20.40	184.8	256
3 / 0	84.9	7 x 3.93	0.39450	11.79	25.70	233.0	296
4 / 0	107	7 x 4.42	0.31188	13.26	32.50	294.7	345
250	126	19 x 2.91	0.26509	14.55	38.80	346.7	384
300	152	19 x 3.19	0.22059	15.95	46.60	416.7	432
350	178	19 x 3.45	0.18860	17.25	52.00	487.3	1020
450	203	19 x 3.69	0.16486	18.45	59.50	557.5	521
500	228	19 x 3.91	0.14683	19.55	66.80	626.0	561
550	253	19 x 4.12	0.13224	20.60	74.20	695.0	600
600	279	37 x 3.10	0.11995	21.70	83.90	766.2	640
650	303	37 x 3.23	0.11049	22.61	91.00	831.9	674
700	330	37 x 3.37	0.10150	23.59	94.90	905.5	712
750	354	37 x 3.49	0.09464	24.43	101.00	971.2	745
800	381	37 x 3.62	0.08796	25.34	109.00	1045.0	781
850	404	37 x 3.73	0.08285	26.12	116.00	1109.0	811
900	456	37 x 3.96	0.07351	27.72	131.00	1250.0	876
1000	508	37 x 4.18	0.06597	29.26	146.00	1393.0	938
1250	631	61 x 3.63	0.05306	32.67	179.00	1732.0	1078
1500	759	61 x 3.98	0.04414	35.82	215.00	2082.0	1210
1750	886	61 x 4.30	0.03781	38.70	251.00	2431.0	1332

The above data is approximate and subject to manufacturing tolerances

Note:-

the size 404 mm2 may be requested 405 mm2 as per customer request with the same above specification
the size 508 mm2 may be requested 506.7 mm2 as per customer request with the same above specification
The above values of current carrying capacity are calculated due to frequency up to 60 Hz, assuming an ambient temperature of 35 °C , final temperature of 80 °C, wind velocity of 0.6 m/sec, solar radiation of 900 W/m2, emissivity coefficient of 0.6, solar radiation absorption coefficient of 0.5

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G - ASTM B 399 (ALCOA American Sizes)							
Akron	15.5	7 x 1.68	2.1590	5.04	4.92	42.6	100
Alton	24.7	7 x 2.12	1.3560	6.36	7.83	67.8	135
Ames	39.2	7 x 2.67	0.8547	8.01	12.40	107.5	181
Azusa	62.4	7 x 3.37	0.5365	10.11	18.90	171.3	243
Anaheim	78.6	7 x 3.78	0.4264	11.34	23.80	215.6	282
Amherst	99.3	7 x 4.25	0.3373	12.75	30.00	272.5	328
Alliance	125	7 x 4.77	0.2678	14.31	37.80	343.2	380
Butte	159	19 x 3.26	0.2112	16.30	46.50	435.1	444
Canton	200	19 x 3.66	0.1670	18.30	58.60	548.5	516
Cairo	236	19 x 3.98	0.1417	19.90	69.20	648.6	1020
Darien	284	19 x 4.36	0.1181	21.80	83.10	778.3	645
Elgin	331	19 x 4.71	0.1012	23.55	97.00	908.3	713
Flint	375	37 x 3.59	0.0894	25.13	107.00	1028.0	772
Gereeley	470	37 x 4.02	0.0713	28.14	135.00	1289.0	893
1077.4	547	37 x 3.38	0.0612	23.66	156.00	1502.0	914
1165.1	590	61 x 3.51	0.0568	31.59	167.00	1620.0	1033
1259.6	638	61 x 3.65	0.0525	32.85	181.00	1751.0	1085
1348.8	685	61 x 3.78	0.0489	34.02	194.00	1878.0	1134
1439.2	729	61 x 3.90	0.0460	35.10	207.00	1999.0	1179

The above data is approximate and subject to manufacturing tolerances

Note:-

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